30554 GNSS – Spring 2026

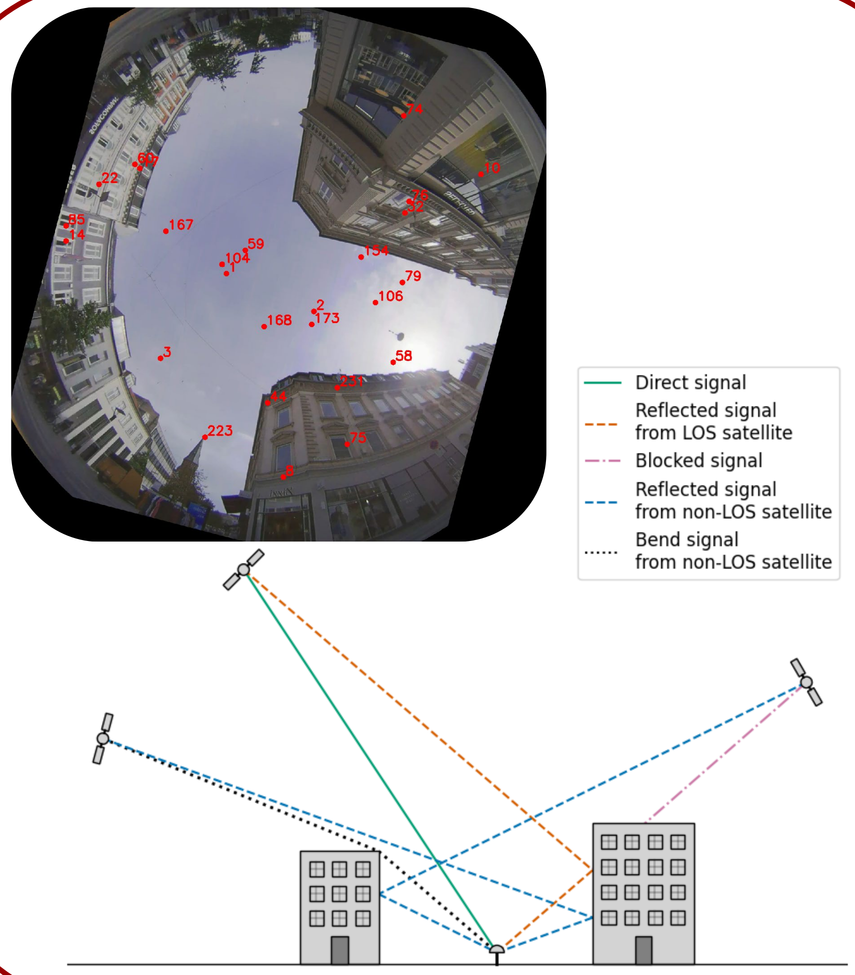
Student projects

Contextual Filtering of Non-LOS satellites

- Project description:

In urban environments, GNSS positioning can be severely degraded by multipath, where signals reflect off buildings before reaching the receiver. A key contributor to this degradation is the inclusion of non-line-of-sight (non-LOS) satellites in the positioning solution. In this project, you will attempt to improve the positioning accuracy by utilizing contextual filtering, using a sky image captured at the measurement site to determine which satellites have a clear LOS and filtering out those that do not.

- Project objectives:
  - Collect data with strong multipath presence
  - Develop a sky-mask from a reference image of the measurement site to filter out non-LOS satellites
  - Estimate multipath errors using a provided equation of each satellite and relate the estimated errors to the geometry of the individual satellites
  - Evaluate the effect of the contextual filtering by comparing positioning accuracy before and after applying the sky-mask



Improved stand-alone positioning

- Project description:

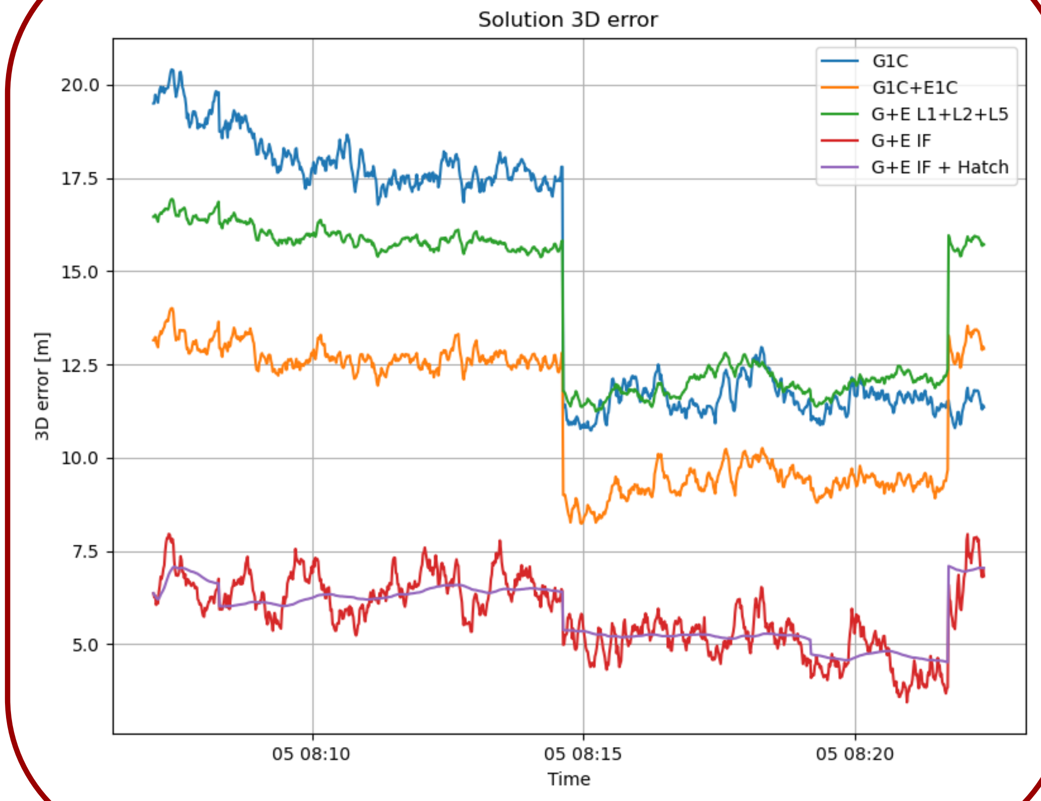
When using our GNSS receiver, we naturally want the best possible accuracy. This project is about investigating the limits of stand-alone positioning, and seeks to answer the question:

What accuracy can we achieve if we only have access to the raw data from our own GNSS receiver?

- Project objectives:

Several things can be implemented in the positioning code as part of this project:

1. Hatch filter
2. Weighted solutions based on elevation or signal strength
3. Multi-system or multi-frequency solutions



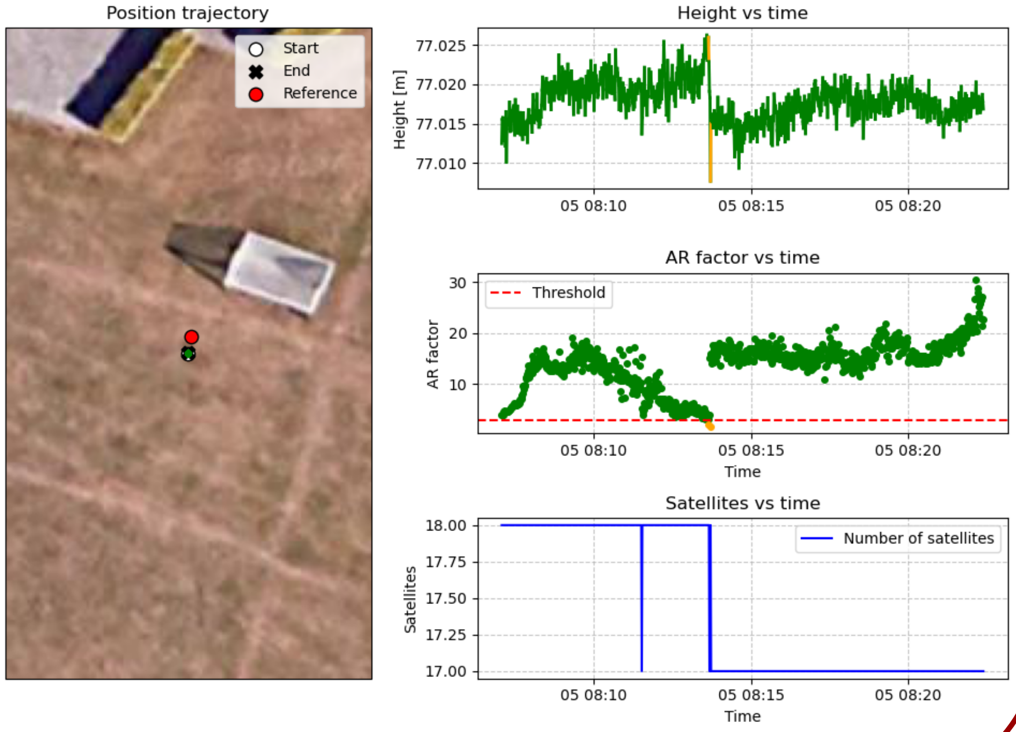
Real-Time Kinematic (RTK) positioning

- Project description:

The best possible accuracy that can be achieved is using the RTK method employing differential carrier phase observations.

The purpose of this project is to try to implement a simple RTK approach on the data from the field work sessions.

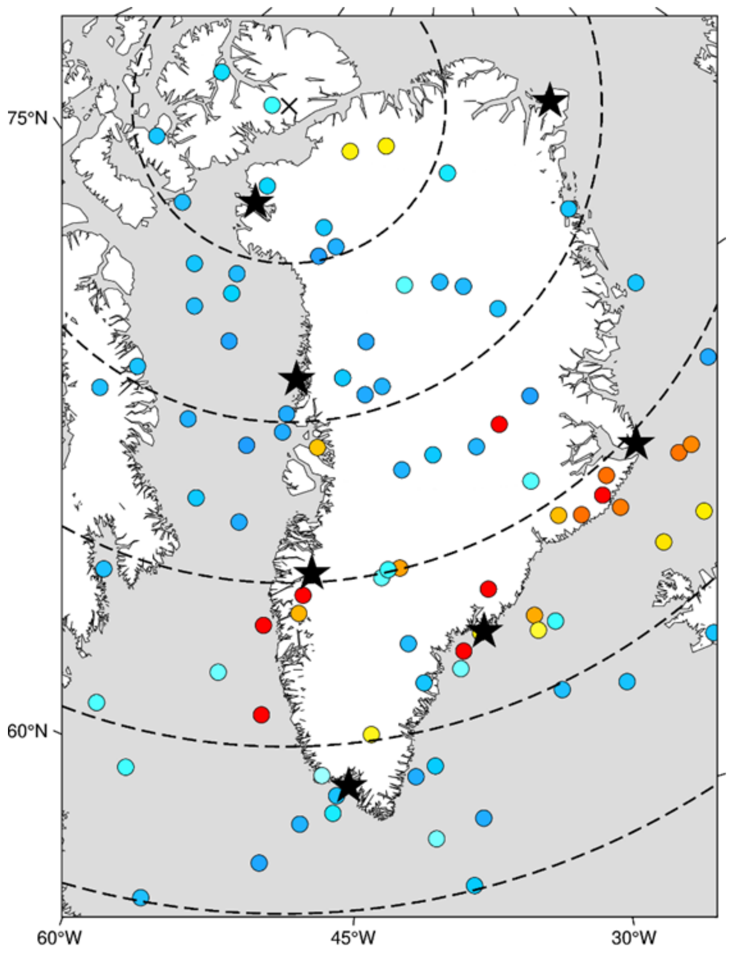
- Project objectives:
  - Implement the LAMBDA method to achieve cm-level positioning accuracy
  - Investigate the impact of the baseline
  - Investigate what further steps can be taken to improve the RTK method



Mapping ionospheric cycle slips

- Project description:

When GNSS signals are affected by space weather, it can affect different line of sights differently. This project focuses on mapping which line of sight are especially impacted by the ionosphere, so we can filter which satellites to use.

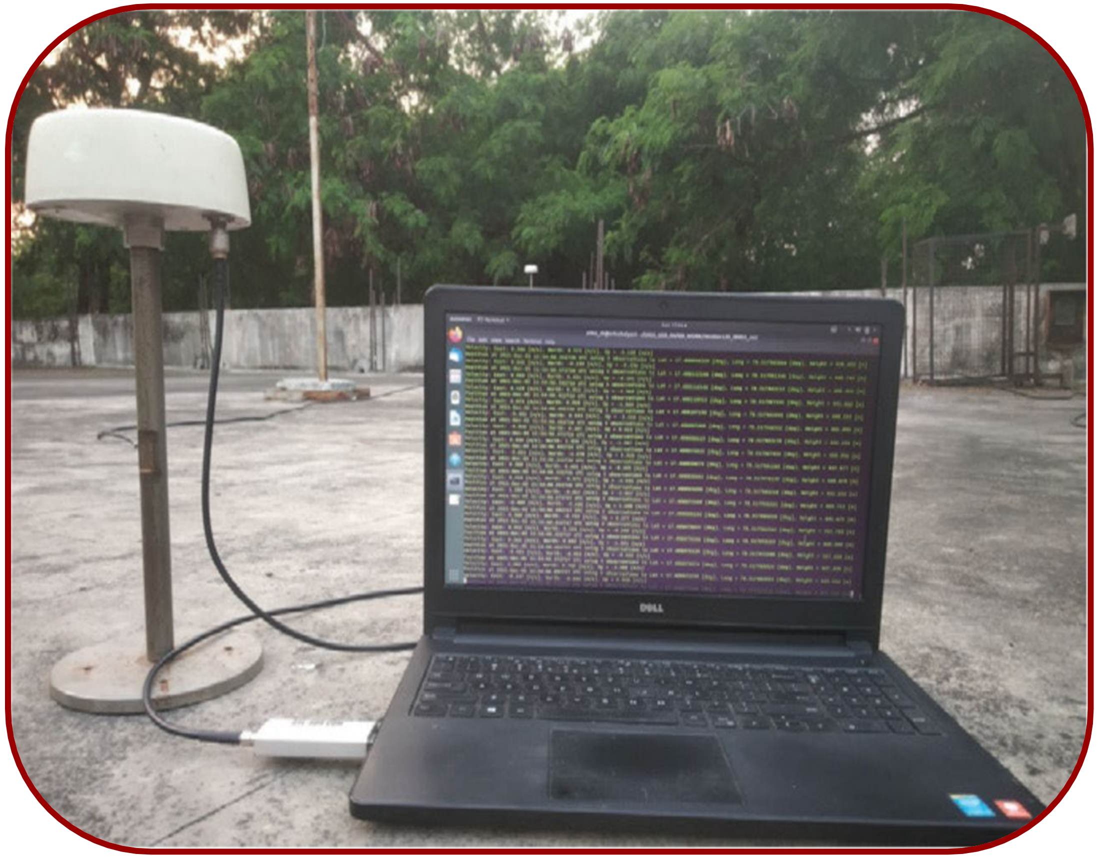


- Project objectives:
  - Identify a storm
  - Implement cycle slip algorithm
  - Map with SP3 files

Exploring parameter selection for a GNSS software receiver

- Project description:
- The settings used in a baseband processor are highly dependent on the operating scenario. For example, chip-spacing may be narrow for applications requiring high accuracy and much looser in cases involving receiver motion or overhead cover.
  This project will explore parameter selections for a GNSS software receiver by collecting data in different scenarios to support experimentation with these settings.

- Project objectives:
  - Make recordings of raw IQ data that supports GNSS processing with a software-defined radio (SDR).
  - Work with GNSS-SDR to obtain position-solutions.
  - Investigate the impact of varying the involved parameters of acquisition and tracking, subject to different scenarios, similar to field tests in lecture 5.



GNSS and INS fusion

- Project description:
- This proposal is about combining GNSS and IMU data in a Loosely Coupled Kalman Filter. GNSS is vulnerable to a number of errors and can also lose reception completely in severe cases. This is especially true in surroundings with many obstacles, such as in an urban setting. In order to bridge these shortcoming GNSS are often combined with Inertial Navigation for ensuring a more robust and accurate solution.

- Project objectives:
  - Perform data collection with UGV
  - Implement the Inertial Navigation Equations and a Loosely coupled Kalman Filter
  - Optimize the performance by tuning key parameters

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